SS Credit: Site Assessment

This credit applies to

- BD+C: New Construction (1 point)
- BD+C: Core and Shell (1 point)
- BD+C: Schools (1 point)
- BD+C: Retail (1 point)
- BD+C: Data Centers (1 point)
- BD+C: Warehouses and Distribution Centers (1 point)
- BD+C: Hospitality (1 point)
- BD+C: Healthcare (1 point)

INTENT

To assess site conditions, environmental justice concerns, and cultural and social factors, before design to evaluate sustainable options and inform related decisions about site design.

REQUIREMENTS

NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare

Complete and document a site survey or assessment[2] that includes the following information:

Topography.

- Contour mapping
- Unique topographic features
- Slope stability risks

Hydrology.

- Special Flood Hazard Areas (SPFHA) as determined by FEMA's Flood Insurance Rate Map (FIRM) (or local equivalent for projects outside the U.S.)
- Delineated natural water bodies wetlands, lakes, streams, and shorelines (refer to U.S. EPA's Clean Water Act or local equivalent for projects outside the U.S.)
- Rainwater collection and reuse opportunities
- Impervious and pervious surfaces within the site boundary

Climate.

- Solar exposure and shading opportunities
- Heat island effect potential
- Seasonal sun angles
- Prevailing winds
- Average monthly precipitation and temperature ranges

Vegetation.

- Primary vegetation types
- Greenfield area
- Significant tree mapping
- Federal or state threatened or endangered species lists; for projects outside the U.S., International Union for Conservation of Nature (IUCN) Red List of Threatened Species
- Invasive plant species listed by regional, state, or federal entities
- EPA Level III ecoregion description (or local equivalent)

[2] Components adapted from the Sustainable Sites Initiative: Guidelines and Performance Benchmarks 2009, Prerequisite 2.1: Site Assessment.

Soils.

- Natural Resources Conservation Service soils delineation (or local equivalent for projects outside the U.S.)
- U.S. Natural Resources Conservation Service (or local equivalent for projects outside the United States) prime farmland, unique farmland, farmland of statewide importance, or farmland of local importance
- Healthy soils
- Previous development
- Disturbed soils

Human use.

- Views
- Adjacent transportation infrastructure, bicycle network, and bicycle storage
- Adjacent diverse uses
- Construction materials with existing recycle or reuse potential

Human health effects.

- Proximity of vulnerable populations
- Adjacent physical activity opportunities
- Proximity to major sources of air and water pollution

The survey or assessment should demonstrate the relationships between the site features and topics listed above and how these features influenced the project design; give the reasons for not addressing any of those topics.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The removal of a U.S.-specific standard, TR-55, from the required considerations and broadening of this concept to account for all impervious and pervious surfaces on site has made this updated credit more approachable to most project teams. The credit now requires projects to describe their EPA Level III Ecoregion (or local equivalent) in order to understand what native and adapted vegetation is appropriate for their site, integrative information that better contextualizes the credit intent.

Further Explanation

Refer to the LEED v4 reference guide, with the following modifications:

- Hydrology: Disregard any mention of the Natural Resources Conservation Service TR-55 program as it is no longer required.
- Hydrology: Estimate the water storage capacity of the site by calculating the area of impervious and pervious surfaces. Table 2 lists of other possible sources of information.
- Vegetation:
    - Source – EPA Level III Ecoregion Descriptions (ftp://newftp.epa.gov/EPADataCommons/ORD/Ecoregions/pubs/NA_TerrestrialEcoregionsLevel3_Final-2june11_CEC.pdf)
    - Description – Descriptions identifying North American ecoregions and detailing their associated ecosystems and vegetation types.

Required Documentation

- Provide a map illustrating the topography of the site.
- Provide a map illustrating the site's Special Flood Hazard Areas (SPFHA) as determined by FEMA's Flood Insurance Rate Map (FIRM) (or local equivalent showing the 100-year floodplain for projects outside the U.S.).
- Provide the description of the site's EPA Level III ecoregion (or local equivalent).

Referenced Standards

- EPA Level III Ecoregion Descriptions: ftp://newftp.epa.gov/EPADataCommons/ORD/Ecoregions/pubs/NA_TerrestrialEcoregionsLevel3_Final-2june11_CEC.pdf

Connection to Ongoing Performance:

- LEED O+M SS credit Site Management: Conducting a site assessment and identifying natural areas providing habitat will fulfill the related credit's site assessment requirements.
- LEED O+M LT credit Transportation Performance: Analyzing the surrounding sites and diverse uses, transportation infrastructure, bicycle network, as well as assessing existing bicycle facilities and potential future facility needs, may help inform and influence the improvement of a project's transportation performance score.
- LEED O+M SS credit Rainwater Management: Studying the climate, rainfall, and hydrology of the site and watershed will help determine applicable strategies to earn the related performance-based credit.
- LEED O+M SS credit Heat Island Reduction: Site assessment can lead to identification of paving, shading, or roofing materials that can contribute to requirements of the related performance-based credit.
- LEED O+M EA credit Energy Performance: An analysis of the climate, including solar access, temperatures, diurnal swings, wind patterns, humidity, and rainfall will support more effective passive and active energy efficiency strategies.